Program 10: Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

Step 1: Initialize Minikube

minikube start --cpus=2 --memory=4096

Step 2: Enable Kubernetes Dashboard

minikube dashboard

Step 3: Create the Deployment File

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 4: Create the Service File

Welcome | deployment.yaml | service.yaml

service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Step 5: Apply Resources

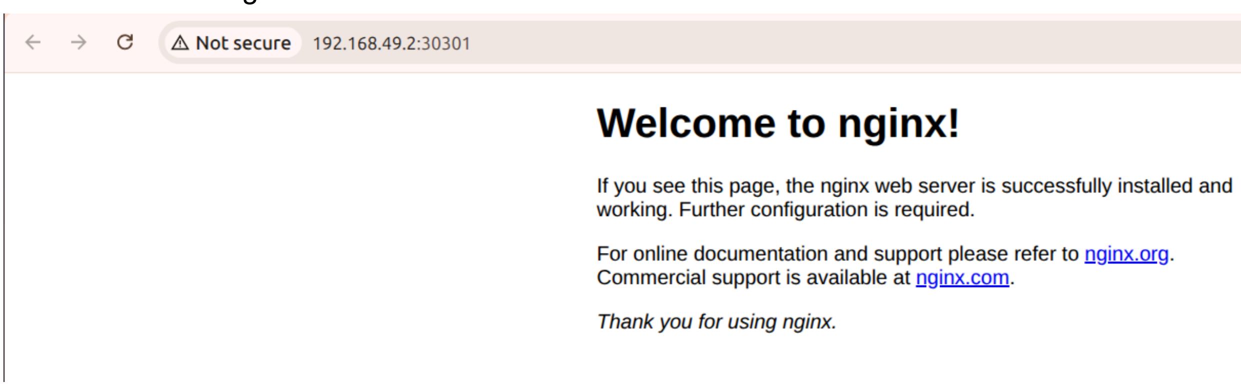
kubectl apply -f deployment.yaml
kubectl apply -f service.yaml

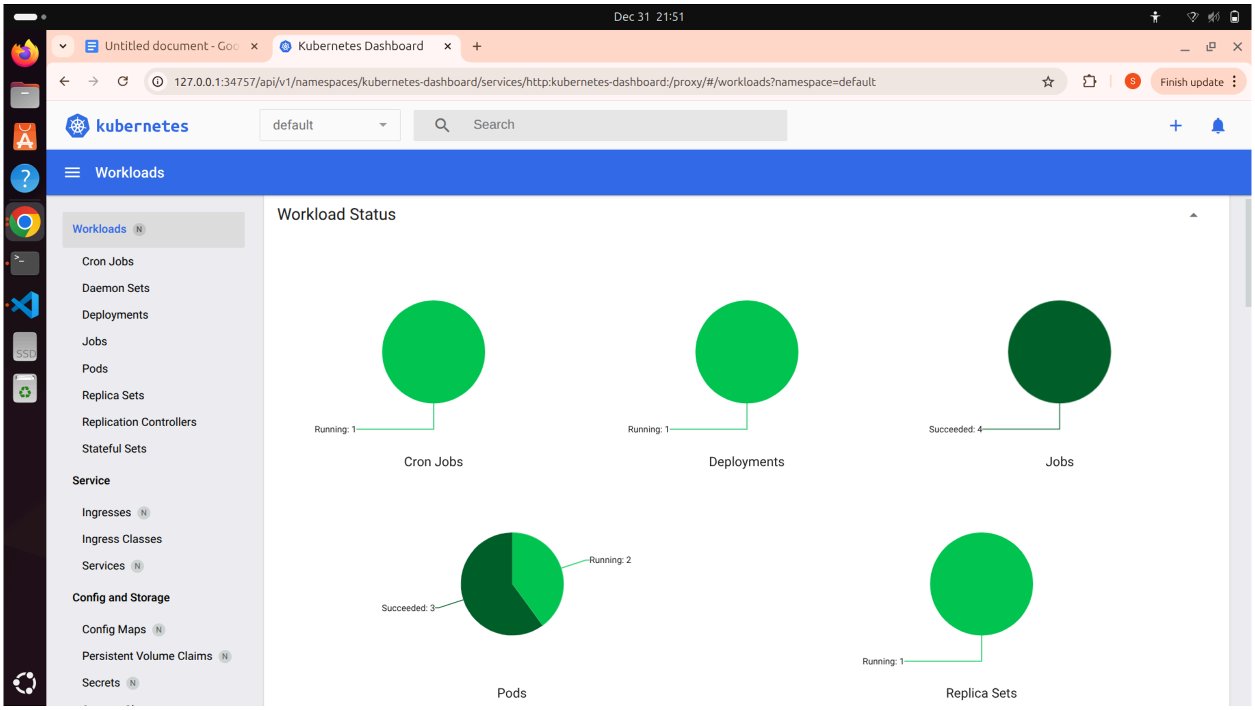
Step 6 – 10: Monitor via CLI

```
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$ kubectl get pods
NAME                               READY   STATUS              RESTARTS      AGE
mycron-29453251-mvs76              0/1     Completed           0             46m
mycron-29453252-lwwbg              0/1     Completed           0             45m
mycron-29453296-tcb76              0/1     Completed           0             24s
mycron-29453297-cq892              0/1     ContainerCreating   0             9s
nginx-deployment-6f9664446b-88j6b  1/1     Running             1 (44m ago)   75m
nginx-deployment-6f9664446b-nssd5  1/1     Running             1 (44m ago)   75m
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   36d   v1.34.0
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$ kubectl get svc
NAME            TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes      ClusterIP   10.96.0.1       <none>        443/TCP        77m
nginx-service   NodePort    10.109.22.119   <none>        80:30301/TCP   76m
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$ kubectl logs nginx-deployment-6f9664446b-88j6b
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/31 16:17:00 [notice] 1#1: using the "epoll" event method
2025/12/31 16:17:00 [notice] 1#1: nginx/1.29.4
2025/12/31 16:17:00 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/31 16:17:00 [notice] 1#1: OS: Linux 6.14.0-33-generic
2025/12/31 16:17:00 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/31 16:17:00 [notice] 1#1: start worker processes
2025/12/31 16:17:00 [notice] 1#1: start worker process 29
2025/12/31 16:17:00 [notice] 1#1: start worker process 30
2025/12/31 16:17:00 [notice] 1#1: start worker process 31
2025/12/31 16:17:00 [notice] 1#1: start worker process 32
2025/12/31 16:17:00 [notice] 1#1: start worker process 33
2025/12/31 16:17:00 [notice] 1#1: start worker process 34
2025/12/31 16:17:00 [notice] 1#1: start worker process 35
2025/12/31 16:17:00 [notice] 1#1: start worker process 36
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$
```

Step 11: Access the Application

minikube service nginx-service





Step 12 – 14: Clean-up Process

kubectl delete pods --all

kubectl delete svc --all

kubectl delete deployments --all

```
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$ kubectl delete pods --all
pod "mycron-29453300-d579v" deleted
pod "mycron-29453301-vg4rp" deleted
pod "mycron-29453302-f6bdf" deleted
pod "nginx-deployment-6f9664446b-88j6b" deleted
pod "nginx-deployment-6f9664446b-nssd5" deleted
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$ kubectl delete svc --all
service "kubernetes" deleted
service "nginx-service" deleted
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted
rv24mc094_sangeethan@sangeetha-n-ThinkPad-X1-Carbon-6th:~/Desktop/Devops_lab/lab10$
```